

Supplementary Material

Pitch–Hue Correspondences: A Quantitative Corpus Analysis of Three Centuries of Mapping Systems

S1. The full 70-system catalogue

Table 1: Table S1. The full 70-system catalogue: name, date, primary reasoning family, retention status, and retention note (mapping type; pitch coverage) or exclusion reason.

System	Date	Family	Status	Note
Ancient Persian System	~700 BCE	F3 Cosmological	retained	pitch-class-to-color; diatonic; 7 keys
Chinese Wuxing Five-Tone/Five-Color System	~4th century BCE onward	F3 Cosmological	retained	pitch-class-to-color; pentatonic; 5 keys
Aristotle / Cureau de la Chambre Interval System	~350 BCE / 1650 CE	F3 Cosmological	discarded	interval-to-color; not pitch-class
Indian Navarasa Color System	~2nd century BCE	F3 Cosmological	discarded	rasa-to-color; not pitch-class
Indian Svvara-to-Color Associations	Traditional, codified ~12th century	F3 Cosmological	retained	pitch-class-to-color; heptatonic (saptak); 1 keys
Al-Kindi's Cosmological Oud-String System	~850 CE	F3 Cosmological	discarded	string/timbre-to-color; not pitch-class
Rudolph of Saint Trond Modal Colors	~1075	F3 Cosmological	discarded	mode-to-color; not pitch-class
Franchino Gaffurio Modal Colors	1492	F3 Cosmological	discarded	mode-to-color; not pitch-class
Gioseffo Zarlino Interval-Color System	1558	F3 Cosmological	discarded	interval-to-color; not pitch-class
Girolamo Cardanus Interval-Color System	1570	F3 Cosmological	discarded	interval-to-color; not pitch-class
Athanasius Kircher Interval-Color System	1646/1650	F3 Cosmological	discarded	interval-to-color; not pitch-class
David Gottlob Diez Note-Color-Planet System	1723	F3 Cosmological	retained	pitch-class-to-color; chromatic (partial); 7 keys
Lorenz Christoph Mizler Newtonian System	1739	F1 Spectral Analogy	retained	pitch-class-to-color; diatonic; 7 keys
Johann Gottlob Krüger System	1743	F1 Spectral Analogy	retained	pitch-class-to-color; diatonic; 7 keys
Leonhard Euler System	~1760	F1 Spectral Analogy	retained	pitch-class-to-color; diatonic; 7 keys
Louis François Henri Lefébure System	1789	F1 Spectral Analogy	retained	pitch-class-to-color; diatonic; 7 keys
Georg Tobias Ludwig Sachs (First Documented Synesthete)	1812	F5 Synesthetic	retained	pitch-class-to-color; partial; 5 keys
D.D. Jameson Chromatic Spectral Mapping	1844	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys
Charles Fourier Inverted Spectral System	1846	F1 Spectral Analogy	retained	pitch-class-to-color; diatonic; 7 keys
François Sudre / Solresol System	1862	F8 Pedagogical	retained	pitch-class-to-color; diatonic; 7 keys
T. Seemann 12-Note System	1881	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys
Bleuler and Lehmann Synesthesia Survey	1881	F5 Synesthetic	discarded	pitch-to-brightness/lightness only; not hue
Bainbridge Bishop Color Organ Spectral System	1893	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys
Théodore Flournoy Synesthesia Cases	1893	F5 Synesthetic	discarded	qualitative case descriptions
Helena Blavatsky / Theosophical System	~1900	F6 Esoteric	retained	pitch-class-to-color; diatonic; 7 keys
Rosicrucian Order System	~1905	F6 Esoteric	retained	pitch-class-to-color; chromatic; 12 keys
Amy Beach Synesthetic Key-Colors	~1910	F5 Synesthetic	retained	pitch-class-to-color; partial (key-based); 8 keys

System	Date	Family	Status	Note
Leonid Sabaneyev Documentation of Scriabin	1911	F2 Circle of Fifths	discarded	pitch-class-to-color; keys not normalizable to 12 pitch classes
Wassily Kandinsky Timbre-Color System	1911	F4 Emotion-Mediated	discarded	string/timbre-to-color; not pitch-class
Roy de Maistre Spectral System	1919	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys
Leon Theremin / Illumovox Continuous System	1922	F1 Spectral Analogy	discarded	continuous/microtone; not discrete pitch-class
Adrian Bernard Klein Spectroscope System	1926	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys
Artists Who Explicitly Rejected Fixed Pitch-Color Mappings	1906–1960s	unclassified	discarded	none; no usable pitch-hue mappings
I.J. Belmont Equal-Division Spectral System	1944	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys
August Aepli Diatonic System	pre-1986	F1 Spectral Analogy	retained	pitch-class-to-color; diatonic (8-note); 7 keys
Galeyev-Vanechkina / Prometei Institute Schema	1975	F2 Circle of Fifths	discarded	key-to-color; keys not normalizable to 12 pitch classes
Steve Zieverink Physical Wavelength Transposition	2004	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys
Jack Ox Virtual Color Organ	1977–present	F2 Circle of Fifths	discarded	key-to-color (with timbre extensions); keys not normalizable to 12 pitch classes
Katherine Lubar Itten Color Wheel Application	2004	F2 Circle of Fifths	discarded	color-interval analogy; not pitch-class mapping
Japanese Gogyo Adopted Wuxing System	6th century CE onward	F3 Cosmological	discarded	pitch-class-to-color (implicit via pentatonic); keys not normalizable to 12 pitch classes
Korean Ohaeng Adopted Wuxing System	Traditional	F3 Cosmological	discarded	pitch-class-to-color (implicit via pentatonic); keys not normalizable to 12 pitch classes
Lawrence Marks Pitch-Brightness Correspondence	1974/1975	F7 Psychophysical	discarded	pitch-to-brightness/lightness only; not hue
Timothy Hubbard Pitch-Lightness Study	1996	F7 Psychophysical	discarded	pitch-to-brightness/lightness only; not hue
Mondloch and Maurer Developmental Pitch-Lightness	2004	F7 Psychophysical	discarded	pitch-to-brightness/lightness only; not hue
Palmer, Schloss, Xu, and Prado-León Emotion-Mediated Study	2013	F4 Emotion-Mediated	discarded	emotion/genre-mediated; not direct pitch-class
Palmer, Langlois, and Schloss Replication	2016	F4 Emotion-Mediated	discarded	emotion/genre-mediated; not direct pitch-class
Hamilton-Fletcher et al. Equiluminant Hue Study	2017	F7 Psychophysical	discarded	pitch-to-hue (controlled)
Whiteford et al. Genre-Dependent Music-Color	2018	F7 Psychophysical	discarded	emotion/genre-mediated; not direct pitch-class
Anikin and Johansson IAT Pitch-Hue Study	2019	F7 Psychophysical	discarded	pitch-to-hue (null result)
Spence and Di Stefano Comprehensive Review	2022	F7 Psychophysical	discarded	review (no novel mappings); no usable pitch-hue mappings
Loconsole et al. Tortoise Pitch-Lightness Study	2025	F7 Psychophysical	discarded	pitch-to-brightness/lightness only; not hue
Sean Day Compiled Synesthesia Databases	2004–2016	F5 Synesthetic	discarded	synesthesia aggregate/individual; not fixed system
Novich, Cheng, and Eagleman Synesthesia Battery	2011	F5 Synesthetic	discarded	synesthesia classification; no usable pitch-hue mappings
Gregersen et al. Absolute Pitch and Synesthesia	2013	F5 Synesthetic	discarded	synesthesia aggregate/individual; not fixed system

System	Date	Family	Status	Note
Friedrich Mahling Historical Bibliography	1926	unclassified	discarded	bibliography; no usable pitch-hue mappings
Neil Harbisson Sonochromatic Scale	2003–present	F1 Spectral Analogy	discarded	continuous/microtone; not discrete pitch-class
Chroma-Notes / Boomwhackers Educational System	~1995	F8 Pedagogical	retained	pitch-class-to-color; chromatic; 7 keys
Figurenotes Accessibility System	1996	F8 Pedagogical	retained	pitch-class-to-color; diatonic; 7 keys
EyeMusic Sensory Substitution Device	2014	F8 Pedagogical	discarded	string/timbre-to-color; not pitch-class
Chromatone Open-Source System	2020s	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys
Fred Collopy 'Three Centuries of Color Scales' Chart	2009	unclassified	discarded	comparative visualization; no usable pitch-hue mappings
Jörg Jewanski 'Ist C = Rot?' Doctoral Dissertation	1999	unclassified	discarded	historical compilation; no usable pitch-hue mappings
Kenneth Peacock 'Instruments to Perform Color-Music'	1988	unclassified	discarded	historical survey of color organs; no usable pitch-hue mappings
José Luis Caivano 'Color and Sound: Physical and Psychophysical Relations'	1994	unclassified	discarded	structural analysis; no usable pitch-hue mappings
Louis Bertrand Castel Ocular Harpsichord	1725	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys; snowball addition
George Field Chromatic Equivalents	1816	F1 Spectral Analogy	retained	pitch-class-to-color; diatonic; 7 keys; snowball addition
Alexander Wallace Rimington Colour-Organ	1895	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys; snowball addition
David A. Wells Proportional Frequency System	1980	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys; snowball addition
Ralph W. Pridmore Wavelength-Frequency Transposition	1992	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys; snowball addition
Soundmap (Mobile Application)	2010s	F1 Spectral Analogy	retained	pitch-class-to-color; chromatic; 12 keys; snowball addition

S2. Family agreement uncertainty and the chance baseline

Table 2: Table S2a. Family agreement scores with bootstrap 95% confidence intervals (10,000 draws, re-samples constrained to at least two distinct systems) and family-label permutation p -values (10,000 permutations; p is the proportion of label shuffles producing a family score $\geq$ the observed one). A global heterogeneity permutation test on the size-weighted variance of family means is non-significant ($T_{\text{obs}} = 0.0015$, $p = .81$).

Family	N	Agreement	95% CI	Perm. p	Null mean (SD)
F8 Pedagogical	3	0.348	[0.301, 0.712]	0.33	0.323 (0.147)
F5 Synesthetic	5	0.276	[0.196, 0.441]	0.40	0.285 (0.100)
F1 Spectral Analogy	24	0.267	[0.229, 0.353]	0.10	0.244 (0.017)
F6 Esoteric	3	0.201	[0.153, 0.260]	0.80	0.324 (0.147)
F3 Cosmological	4	0.187	[0.178, 0.720]	0.91	0.298 (0.118)

Table 3: Table S2b. Simulated chance distribution of the agreement score: ensembles of n systems of 12 colors drawn uniformly from the sRGB gamut, passed through the identical CIEDE2000 pipeline (2,000 ensembles per n).

Ensemble size n	Mean agreement	95% interval
3	0.227	[0.151, 0.408]
5	0.217	[0.172, 0.310]
15	0.212	[0.192, 0.240]
24	0.212	[0.196, 0.231]

Sensitivity check (31 retained systems). Recomputing the flat analyses on all 31 retained systems in the final catalogue yields a mean per-pitch agreement of 0.237 (range 0.201–0.310), Rayleigh $R = 0.60$, and a semitone–hue distance correlation of $r = 0.49$ ($p = 3.1e - 05$), leaving all conclusions unchanged.

S3. Pedagogical consensus versus canonical spectral order

To test whether the Pedagogical family's convergence simply reuses rainbow order, the F8 consensus hues were compared against the canonical spectral-order mapping (equally spaced hues in chromatic pitch order). Over the seven pitch classes shared by all three F8 systems, the mean angular distance is 54.1° ; allowing a free global rotation of the spectral template (hue origin is arbitrary), the minimum mean distance is 35.1° . For calibration, the Spectral Analogy family (F1), whose systems explicitly encode rainbow order, scores 39.9° unrotated and 35.8° after optimal rotation over all 12 pitch classes. The F8 consensus is therefore about as close to spectral order (after rotation) as the self-declared spectral family itself: the pedagogical convergence is spectrally anchored, consistent with the reading in the main text that educators converge on a spectrally ordered, affectively comfortable mapping rather than on an arbitrary shared palette.

S4. Sentence-transformer probing grids

Table 4: Table S4a. Full 3 (model) $\times$ 6 (descriptor level) probing grid, primary (LLM-generated, color-blocklisted) descriptor set. p -values from condition-specific label-shuffle permutation; q -values Benjamini–Hochberg corrected across the 18 cells. Bold: $p < .05$ before correction.

Model	Descriptor level	Geometry ρ	Perm. p	q_{BH}
MiniLM	Bare names	-0.115	0.942	0.973
MiniLM	Musical context	-0.039	0.590	0.817
MiniLM	Perceptual	-0.040	0.631	0.817
MiniLM	Timbral	-0.156	0.973	0.973
MiniLM	Connotative	-0.049	0.645	0.817
MiniLM	Solfège	0.019	0.344	0.774
MPNet	Bare names	0.055	0.226	0.582
MPNet	Musical context	0.156	0.060	0.251
MPNet	Perceptual	0.203	0.048	0.251
MPNet	Timbral	-0.130	0.916	0.973
MPNet	Connotative	0.124	0.124	0.373
MPNet	Solfège	0.000	0.441	0.794
Nomic	Bare names	0.173	0.070	0.251
Nomic	Musical context	-0.032	0.516	0.817
Nomic	Perceptual	0.269	0.044	0.251
Nomic	Timbral	0.368	0.007	0.132
Nomic	Connotative	-0.060	0.681	0.817
Nomic	Solfège	0.005	0.402	0.794

Table 5: Table S4b. The same grid for the factual descriptor set (deterministically derived from music-theory constants). Bold: $p < .05$ before correction.

Model	Descriptor level	Geometry ρ	Perm. p	q_{BH}
MiniLM	Bare names	-0.115	0.940	0.940
MiniLM	Musical context	-0.036	0.573	0.737
MiniLM	Perceptual	0.164	0.033	0.155
MiniLM	Timbral	0.265	0.048	0.174
MiniLM	Connotative	-0.128	0.850	0.900
MiniLM	Solfège	0.049	0.249	0.499
MPNet	Bare names	0.055	0.221	0.497
MPNet	Musical context	0.334	0.012	0.139
MPNet	Perceptual	-0.004	0.475	0.701
MPNet	Timbral	0.197	0.034	0.155
MPNet	Connotative	-0.081	0.722	0.835
MPNet	Solfège	0.230	0.015	0.139
Nomic	Bare names	0.173	0.058	0.175
Nomic	Musical context	0.109	0.129	0.331
Nomic	Perceptual	-0.008	0.481	0.701
Nomic	Timbral	-0.011	0.506	0.701
Nomic	Connotative	-0.031	0.502	0.701
Nomic	Solfège	-0.089	0.742	0.835

S5. EPA Procrustes statistics

Table 6: Table S5. EPA Procrustes statistics. Observed disparity = 0.559; permutation null (10,000 shuffles of the correspondence assignment): mean = 0.833, SD = 0.106; permutation $p = 0.027$. The sensorimotor configuration is the 3-component PCA (68.3% of variance) of the 11-dimensional profiles of the 10 aligned dimensions. Dimension-wise Pearson correlations between matched coordinates after alignment are given below.

EPA dimension	<i>r</i>	<i>p</i>
Evaluation	0.74	0.014
Potency	0.47	0.166
Activity	0.37	0.293

S6. The coded CMC database

Table 7: Table S6. The coded database of 34 documented cross-modal correspondence pairings: direction, literature-derived effect size (Cohen’s d , with plausible range), mechanism proposed in the source literature, and key sources.

Pairing	Dir.	d [range]	Mechanism	Key sources
pitch height → lightness	+	1.00 [0.80, 1.20]	structural	Marks1974; Marks1987; Ward2006; Spence2011
pitch height → spatial elevation	+	0.80 [0.60, 1.00]	semantic	Spence2011; Rusconi2006
pitch height → size	-	0.65 [0.50, 0.80]	statistical	Mondloch2004; Spence2011
pitch height → angularity	+	0.50 [0.30, 0.70]	structural	Parise2012
pitch height → hue	?	0.20 [0.05, 0.35]	unknown	Ward2006; Whiteford2018
loudness → brightness	+	0.40 [0.20, 0.60]	structural	Marks1987
loudness → size	+	0.60 [0.40, 0.80]	statistical	Smith1992
tempo → saturation	+	0.50 [0.30, 0.70]	emotion	Palmer2013
tempo → lightness	+	0.50 [0.30, 0.70]	emotion	Palmer2013; Collier2001
mode major → hue warmth	+	0.60 [0.40, 0.80]	emotion	Palmer2013
spectral centroid → color temperature	+	0.45 [0.25, 0.65]	structural	Giannakis2001; Palmer2013
consonance → pleasantness	+	0.55 [0.35, 0.75]	emotion	Lahdelma2016
sweetness → roundness	+	0.65 [0.45, 0.85]	statistical	Spence2011
bitterness → angularity	+	0.55 [0.35, 0.75]	statistical	Spence2011
sourness → angularity	+	0.40 [0.20, 0.60]	statistical	Velasco2016
taste intensity → saturation	+	0.50 [0.30, 0.70]	structural	Spence2011
saltiness → whiteness	+	0.35 [0.20, 0.50]	semantic	Wan2014
umami → brownness	+	0.30 [0.15, 0.45]	semantic	Spence2011
smoothness → lightness	+	0.40 [0.20, 0.60]	structural	Ludwig2013
hardness → loudness	+	0.40 [0.20, 0.60]	statistical	Walker2012
roughness → spectral roughness	+	0.50 [0.30, 0.70]	structural	Guest2002
temperature warm → hue warmth	+	0.45 [0.25, 0.65]	semantic	Deroy2013
heaviness → pitch height	-	0.35 [0.15, 0.55]	statistical	Walker2012
odor pleasantness → lightness	+	0.40 [0.25, 0.55]	hedonic	Kemp1997
odor intensity → saturation	+	0.38 [0.18, 0.58]	hedonic	Gilbert2008
citrus → yellow	+	0.42 [0.22, 0.62]	semantic	Schifferstein1999
mint → green	+	0.36 [0.20, 0.52]	semantic	Schifferstein1999
pitch direction → brightness	+	0.35 [0.15, 0.55]	structural	Collier2001
rhythmic complexity → angularity	+	0.30 [0.15, 0.45]	emotion	Palmer2013
timbre brightness → hue warmth	+	0.44 [0.24, 0.64]	structural	Palmer2013
brightness → loudness	+	0.28 [0.12, 0.44]	structural	Marks1987
saturation → taste intensity	+	0.46 [0.26, 0.66]	statistical	Spence2011
redness → sweetness	+	0.41 [0.21, 0.61]	semantic	Spence2015
greenness → sourness	+	0.37 [0.17, 0.57]	semantic	Velasco2016

S7. Consolidated multiplicity table

Table 8: Table S7. Consolidated multiplicity table: all primary and battery-level tests with Benjamini–Hochberg FDR-corrected q -values.

Test	p	q_{BH}
Semitone-hue correlation H2 (permutation)	0.0005	0.005
Interval-variance correlation H_b6	0.0014	0.007
Timbral embedding alignment (Nomic, permutation)	0.0073	0.024
Rayleigh non-uniformity (Monte Carlo)	0.0103	0.026
EPA Procrustes alignment (permutation)	0.0268	0.045
Effect size vs sensorimotor distance	0.0269	0.045
Agreement-consistency H_b9	0.0325	0.046
Family heterogeneity: F1 label-permutation	0.1011	0.126
Family heterogeneity: F8 label-permutation	0.3330	0.370
Family heterogeneity: F5 label-permutation	0.3973	0.397

S8. Mean hue distance by musical interval

Table 9: Table S8. Mean hue distance (degrees on the colour wheel) by musical interval, pooled across all 25 retained systems. Circular pitch-class distance makes the perfect fourth and perfect fifth equivalent (both 5 semitones); their rows are identical by construction.

Interval	Semitones	Mean hue dist. (°)	SD (°)
Minor second	1	36.9	44.6
Major second	2	53.3	44.3
Minor third	3	59.1	38.3
Major third	4	70.5	34.9
Perfect fourth	5	80.7	26.1
Tritone	6	85.7	22.0
Perfect fifth	7	80.7	26.1

S9. Corpus-level hypothesis battery (full 10 tests)

Table 10: Table S9. Corpus-level hypothesis test results. The Account column attributes each hypothesis to the theoretical account it operationalises; only the Statistical rows bear directly on the environmental co-occurrence account.

Hyp.	Description	Account	Result	Key statistic
H_{b1}	Lightness increases with chromatic pitch order	Structural	Fail	$r = -.53, p = .076$
H_{b2}	Agreement is higher for high-chroma assignments	Hedonic	Fail	$r = .55, p = .063$
H_{b3}	Consonant intervals show larger hue distance than dissonant	Structural	Fail	$t = 1.05, p = .34$
H_{b4}	Hue distribution non-uniform	Statistical	Pass	Rayleigh $R = 0.66, p = .003$
H_{b5}	Sharps differ from naturals in lightness or hue	Cultural	Fail	$p = .53 / .95$
H_{b6}	Interval size predicts between-system variance	Statistical	Pass	$r = -.94, p = .001$
H_{b7}	Perfect fifth < tritone hue distance	Statistical	Inconclusive	$80.7^\circ < 85.7^\circ$ (untested)
H_{b8}	Primary triad (C–E–G) hues $\sim 120^\circ$ apart	Structural	Fail	mean 66.4° , dev. 53.6°
H_{b9}	Agreement correlates with consistency	Statistical	Pass	$r = -.62, p = .03$
H_{b10}	Tradition-specific fifths differ from pooled	Cultural	Fail	$t = 0.15, p = .89$

S10. Per-family harmonic structure

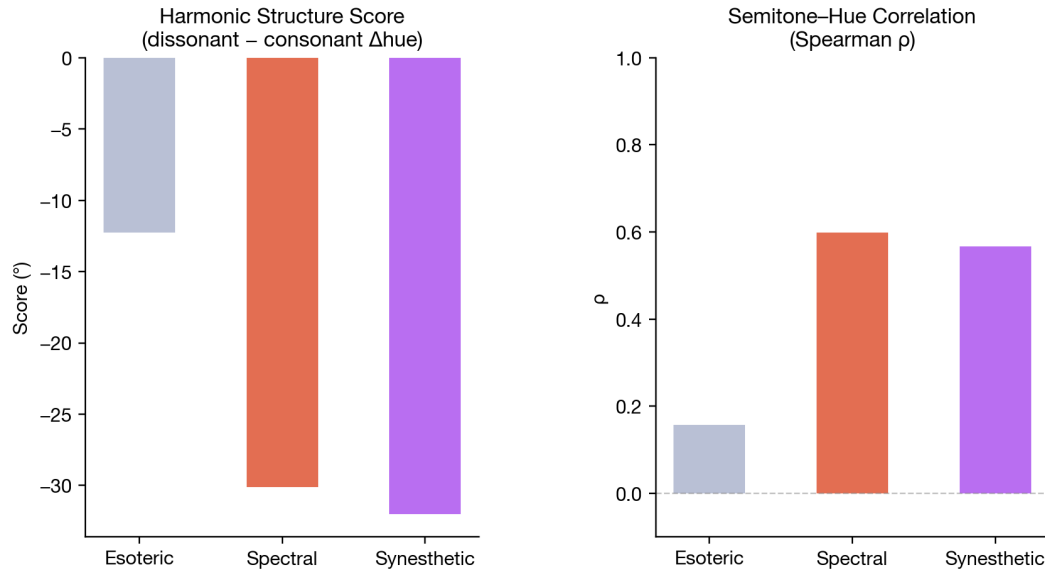


Figure 1: Figure S10. Per-family harmonic structure analysis for three families (Esoteric F6, Spectral Analogy F1, Synesthetic F5). Left: harmonic structure score (mean hue distance for dissonant minus consonant intervals); all three families assign more similar hues to consonant than dissonant intervals. Right: Spearman ρ between semitone distance and mean hue distance; Spectral Analogy and Synesthetic families show substantially stronger interval-to-hue structure ($\rho \approx 0.6$) than the Esoteric family.

S11. Structural theory pairwise alignment heatmap

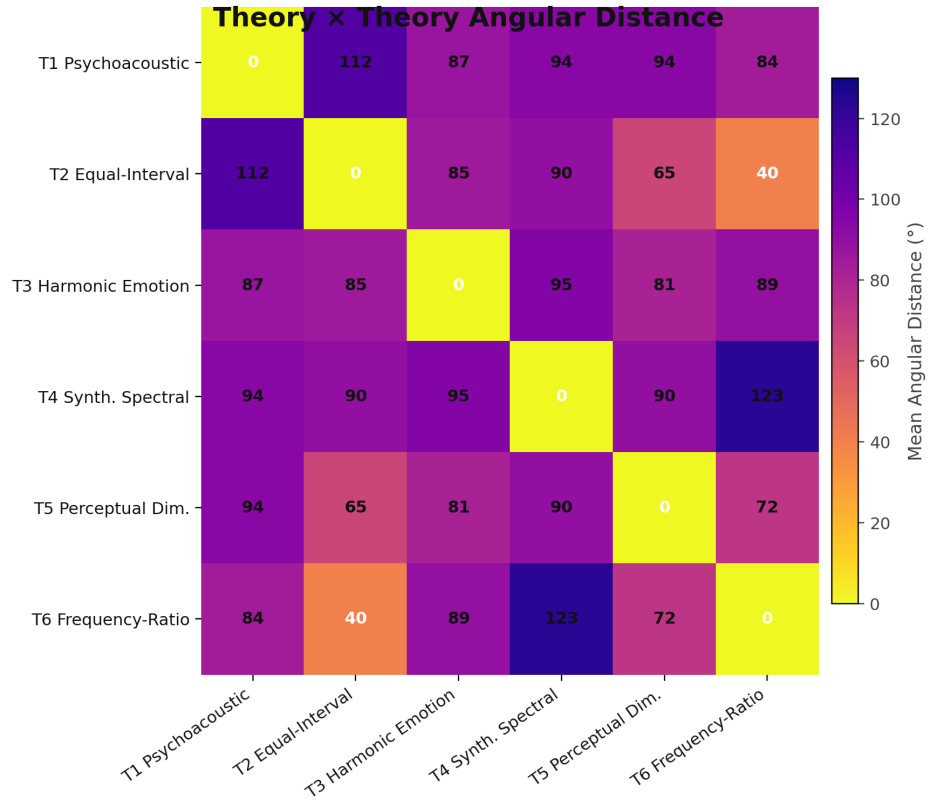


Figure 2: Figure S11. Pairwise alignment heatmap for the six structural theories. Cell colour encodes mean angular distance between predicted hue assignments. IRP and FRI form the tightest pair (40.1°), confirming that they capture the same structural intuition through different mathematical formalisms.

S12. Structural theory predicted hue assignments

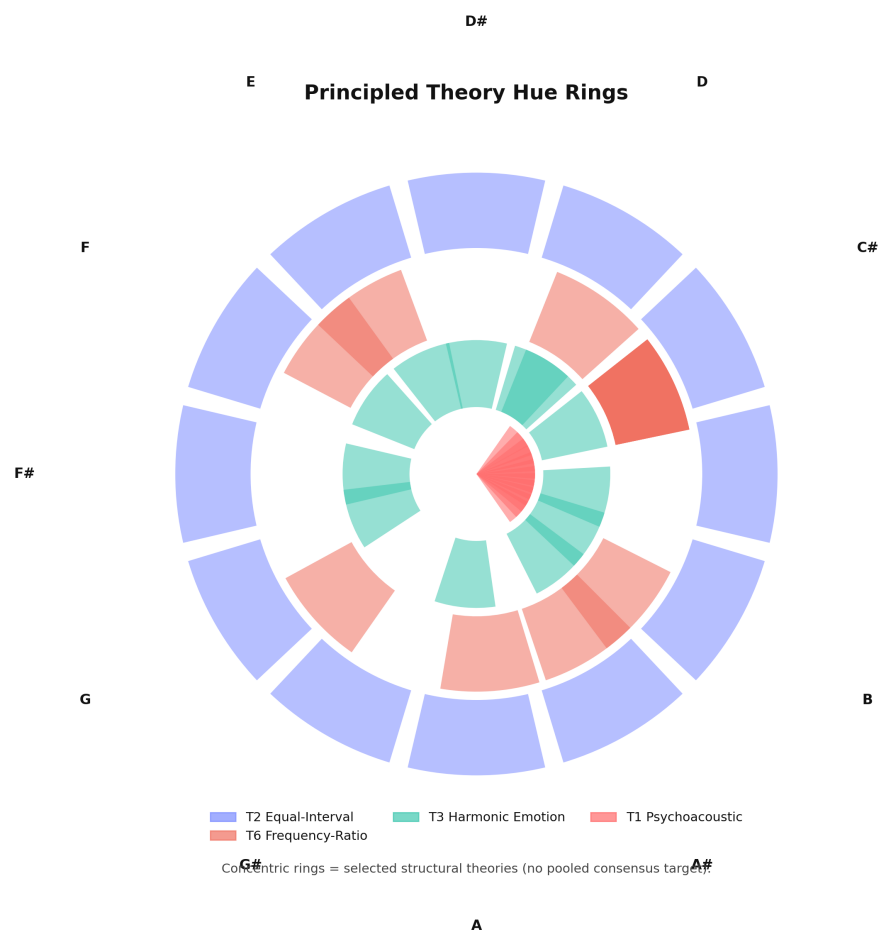


Figure 3: Figure S12. Predicted hue assignments for the six structural theories overlaid on the global consensus wheel. IRP is the closest match to the observed historical consensus, though it diverges substantially in the middle register (D–G#).

S13. Per-model language-model predicted hue maps

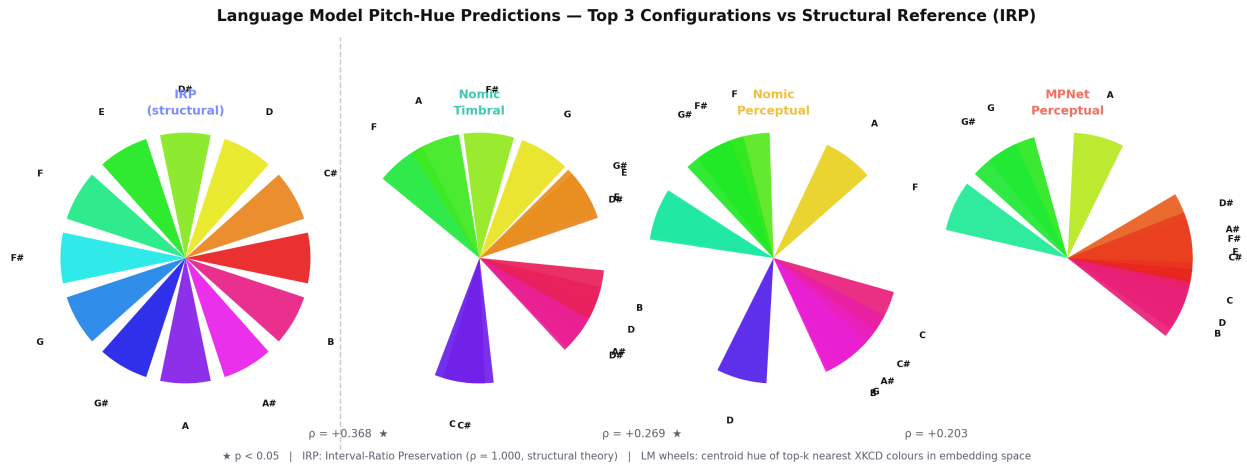


Figure 4: Figure S13. Predicted hue assignments from the three language models (MiniLM, MPNet, Nomic) under LLM-generated timbral descriptors. Nomic shows the most structured prediction (warm reds at lower pitches, cooler hues at higher pitches), consistent with its higher geometry alignment ($\rho = .368$, $p = .007$).

S14. Cross-linguistic and emotion-mediation control panels

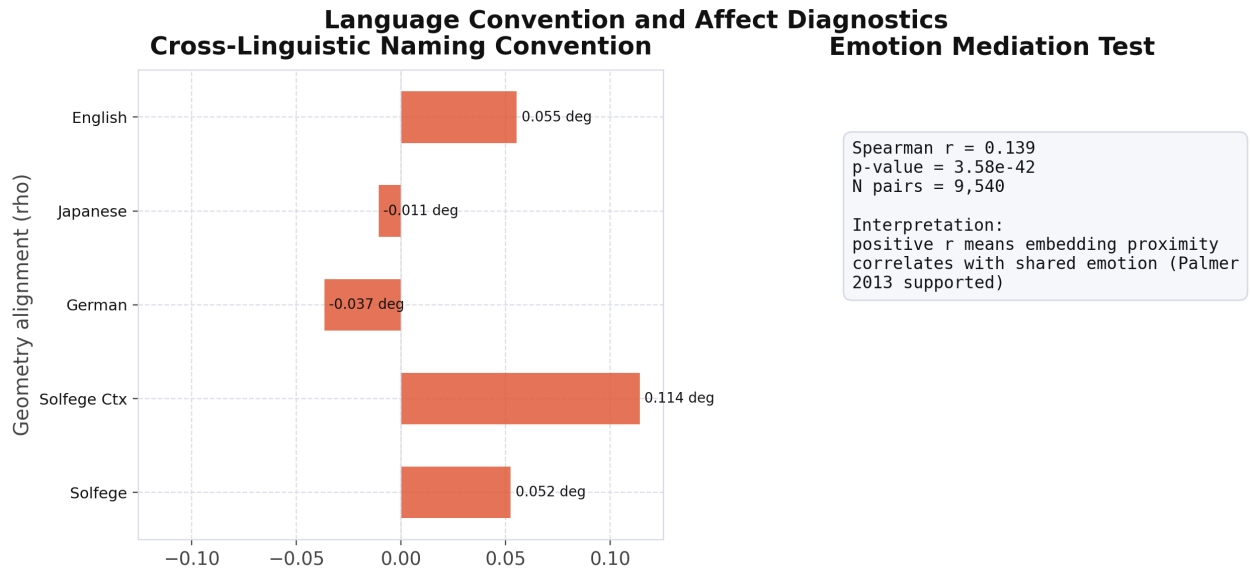


Figure 5: Figure S14. Left: geometry alignment across five naming conventions (English, Japanese, German, solfège with context, bare solfège); pitch–hue geometry is weak across all naming conventions, arguing against name-specific artefacts. Right: the small positive Spearman correlation ($\rho = .139$, $p < .001$) between pitch–colour embedding proximity and shared emotional valence is consistent with the hedonic account.

S15. Complete per-system mapping panel by reasoning family

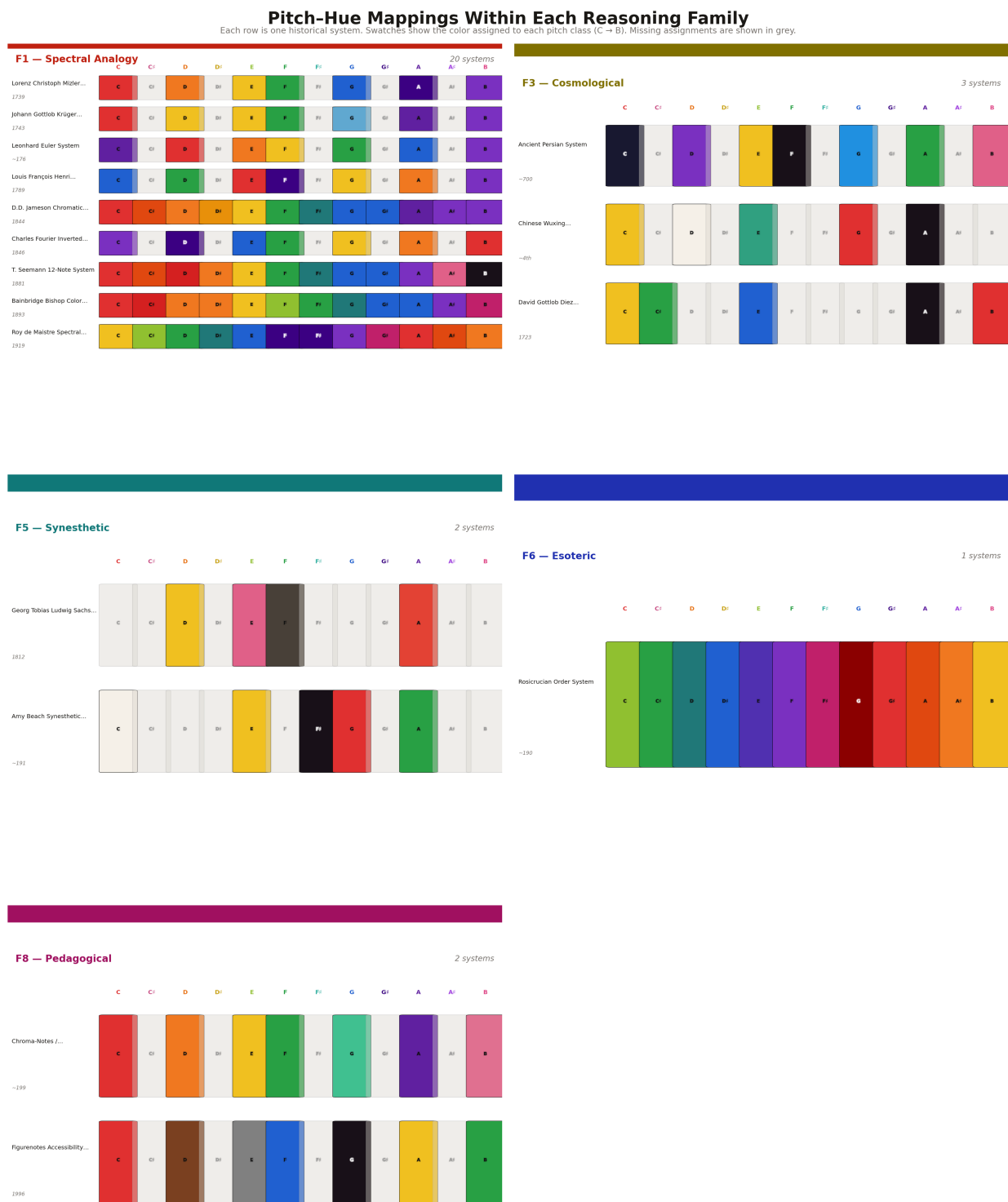


Figure 6: Figure S15. Within-family mapping panel. Each row shows one historical system’s colour assignments for all 12 pitch classes (C to B), grouped by reasoning family (F1–F8). Spectral Analogy (F1) is dominated by rainbow order; Synesthetic (F5) tends toward warm hues at low pitches and cool hues at high; Pedagogical (F8) shows the tightest within-family clustering.